Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

You are given a linear classifier with weight matrix

$$W = \begin{bmatrix} 3 & -1 & -2 \\ -3 & 1 & 4 \end{bmatrix}$$

The classifier takes an input vector: $x = [a,\ b,\ c]$

and computes the class scores by performing the multiplication of matrix W and vector x.

$$Wx = \begin{bmatrix} \text{'Wob Score'} \\ \text{'Bob Score'} \end{bmatrix}.$$

Consider the specific point: $x = [1,\ 3,\ -1]$

Based on the computed scores from Wx, into which class will this point be classified?

A. Wob, because the Wob score is higher than the Bob score

B. Bob, because the Bob score is higher than the Wob score

C. Wob, because both scores are equal

D. Bob, because both scores are equal

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 Multiple Select Type Award: 1 Penalty: 0 Machine Learning

For the binary cross-entropy loss:

$$L(y, \hat{y}) = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$$

where $y \in \{0, 1\}$ is the true label and $\hat{y}$ is the predicted probability.
Which of the following statements is/are correct?

A. The loss can be negative if $\hat{y}$ is close to 1.

B. The loss is always non-negative; it becomes zero only when $\hat{y}$ matches the true label y.

C. The loss grows very large when the predicted probability $\hat{y}$ is close to the wrong class.

D. The loss is always greater than 1 for any prediction.

Your Answer: Correct Answer: B;C Not Attempted Time taken: 00min 00sec Discuss

Q #3 Multiple Select Type Award: 1 Penalty: 0 Machine Learning

Consider a binary logistic regression model **without bias**, where the input vector lies in a 2-dimensional space, i.e., $x = (x_1, x_2) \in \mathbb{R}^2$. The class probabilities are defined as

$$P(y = +1 \mid x) = \sigma(w^\top x), \qquad P(y = -1 \mid x) = 1 - \sigma(w^\top x),$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function.
Which of the following statements is/are correct?

A. $P(-1 \mid (+1, +1)) = 1 - P(+1 \mid (+1, +1))$
B. $P(-1 \mid (+1, +1)) = P(+1 \mid (-1, -1))$
C. $P(-1 \mid (+1, +1)) = P(-1 \mid (-1, -1))$
D. $P(+1 \mid (+1, +1)) = P(+1 \mid (-1, -1))$

Your Answer: Correct Answer: A;B Not Attempted Time taken: 00min 00sec Discuss

Q #4 Multiple Choice Type Award: 1 Penalty: 0.33 Machine Learning

We are given a binary logistic regression model with parameter vector $w \in \mathbb{R}^d$. For any input vector x, the model predicts the probability of class $y = 1$ as:

$$P(y = 1 \mid x) = \frac{1}{1 + \exp(-w \cdot x)}.$$

We want to determine all points x for which this predicted probability becomes exactly $\frac{3}{4}$.

In other words, find the equation of the surface (the set of all such x) that satisfies:

$$P(y = 1 \mid x) = \frac{3}{4}.$$

This surface corresponds to the probability level 0.75.

Which of the following equations correctly represents that boundary?

A. $w \cdot x = 0$
B. $w \cdot x + \log(3) = 0$
C. $w \cdot x + \log\left(\frac{1}{3}\right) = 0$
D. $w \cdot x - \log(5) = 0$

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 00sec Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Mira is training a softmax regression model to classify Chinese antiques into four dynasties: **Tang**, **Song**, **Ming**, and **Qing**. The model outputs a probability distribution over these four classes, and the chosen loss function is the **cross-entropy loss**.

Suppose an antique truly belongs to the **Ming** dynasty. Each option below shows a predicted probability distribution from the model. The probabilities are always listed in the fixed order:

(Tang, Song, Ming, Qing)

Determine which predicted probability distribution will produce the smallest loss.

A. $(0.25, 0.20, 0.30, 0.25)$
B. $(0.01, 0.01, 0.44, 0.54)$
C. $(0.25, 0.25, 0.25, 0.25)$
D. $(0.97, 0.01, 0.01, 0.01)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

You are training a logistic regression model with binary input features $x_1, x_2, \ldots, x_d \in \{0, 1\}$. The prediction is:

$$\hat{y} = \sigma\left(\sum_{j=1}^{d} \theta_j x_j + \theta_0\right), \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

Suppose we augment the dataset by adding the complement of each feature, i.e., for every feature x_j we also add $1 - x_j$ as a new feature.

How will this change the model's ability to fit the training data?

A. The model can now fit the training data better.

B. The model's ability to fit the training data remains the same.

C. The model will fit the training data worse.

D. The model may sometimes improve and sometimes worsen, depending on the dataset.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 11sec | Discuss

Q #7 **Multiple Choice Type** **Award: 1** **Penalty: 0.33** **Machine Learning**

Suppose instead of using one sigmoid in logistic regression, we use two sigmoids in sequence. That is, instead of

$$\hat{y} = \sigma\left(\sum_{j=1}^{d} \theta_j x_j\right),$$

we define

$$\hat{y} = \sigma\left(\sigma\left(\sum_{j=1}^{d} \theta_j x_j\right)\right).$$

What will be the effect on the model's predictions?

A. The model can now learn better since two sigmoids add more nonlinearity.

B. The model behaves the same as the original logistic regression.

C. The model performs worse because the output is squeezed into a narrow range $[0.5, 0.73]$.

D. The effect is uncertain and depends on the dataset.

Your Answer: **Correct Answer: C** **Not Attempted** **Time taken: 00min 00sec** **Discuss**

Q #8 **Multiple Choice Type** **Award: 1** **Penalty: 0.33** **Machine Learning**

Suppose we first fit a LINEAR REGRESSION model and obtain predictions $\hat{y}^{(i)}$ for each training example $x^{(i)}$.

Now, we form a new dataset where each example is $\left[x_1^{(i)}, x_2^{(i)}, \ldots, x_d^{(i)}, \hat{y}^{(i)}\right]$, i.e., we add the model's prediction as an extra feature.

We then fit another LINEAR REGRESSION model on this new dataset.

How will this affect the model's ability to fit the training data?

A. It improves the model's ability to fit since $\hat{y}^{(i)}$ adds a new informative feature.

B. It remains the same because $\hat{y}^{(i)}$ is just a linear combination of the original features.

C. It makes the model worse, because adding $\hat{y}^{(i)}$ introduces redundancy.

D. The effect is unpredictable; sometimes it improves and sometimes it worsens.

Your Answer: **Correct Answer: B** **Not Attempted** **Time taken: 00min 00sec** **Discuss**

Q #9 **Multiple Choice Type** **Award: 1** **Penalty: 0.33** **Machine Learning**

Suppose we first fit a LOGISTIC REGRESSION model and obtain predictions $\hat{y}^{(i)}$ for each training example $x^{(i)}$.

Now, we form a new dataset where each example is $\left[x_1^{(i)}, x_2^{(i)}, \ldots, x_d^{(i)}, \hat{y}^{(i)}\right]$, i.e., we add the model's prediction as an extra feature.

We then fit another LOGISTIC REGRESSION model on this new dataset.

How will this affect the model's ability to fit the training data?

A. It improves the model's ability to fit since $\hat{y}^{(i)}$ is a nonlinear transformation of the original features.

B. It remains the same because $\hat{y}^{(i)}$ does not add any new information.

C. It makes the model worse, because adding $\hat{y}^{(i)}$ confuses the second model.

D. The effect is unpredictable; sometimes it improves and sometimes it worsens.

Your Answer: **Correct Answer: A** **Not Attempted** **Time taken: 00min 00sec** **Discuss**

Q #10 **Multiple Choice Type** **Award: 1** **Penalty: 0.33** **Machine Learning**

You train a logistic regression classifier with ridge regularization.

Suppose you use all 100 data points labeled $+1$ and only one data point labeled -1 (at $x = [-2, 2]^T$) as the training dataset. The remaining data points are used as the testing dataset.

Which of the following will the classifier reliably achieve?

A. Low training error

B. Low testing error

C. Both A and B

D. Neither A nor B

Your Answer: **Correct Answer: A** **Not Attempted** **Time taken: 00min 00sec** **Discuss**

Q #11 Multiple Choice Type Award: 2 Penalty: 0.67 Machine Learning

Suppose we fit two logistic regression models:

- Model 1 uses the standard sigmoid function

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

- Model 2 uses the base-2 sigmoid function

$$\sigma_2(z) = \frac{1}{1+2^{-z}}$$

Which statement is correct?

A. Model 1 can fit the training data better than Model 2.

B. Model 2 can fit the training data better than Model 1.

C. Both models are equally expressive.

D. Either Model 1 or Model 2 may perform better depending on the dataset.

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 00sec Discuss

Q #12 Multiple Select Type Award: 2 Penalty: 0 Machine Learning

In softmax regression, the predicted probability for class j is

$$p_j = \frac{e^{z_j}}{\sum_{k=1}^{K} e^{z_k}}$$

where z_j is the score of class j. The derivative of p_t (probability of the true class t) with respect to the scores is

$$\frac{\partial p_t}{\partial z_j} = \begin{cases} p_t\left(1 - p_t\right), & j = t \\ -p_t p_j, & j \neq t \end{cases}$$

For the true class t, the cross-entropy loss is

$$L = -\log(p_t)$$

Which of the following correctly describes $\frac{\partial L}{\partial z_j}$?

A. If $j = t$ (true class): $\frac{\partial L}{\partial z_t} = p_t - 1$

B. If $j \neq t$ (wrong class): $\frac{\partial L}{\partial z_j} = p_j$

C. If $j = t$: $\frac{\partial L}{\partial z_t} = p_t$

D. If $j \neq t$: $\frac{\partial L}{\partial z_j} = p_j - 1$

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider a labeled set of examples from three classes ($K = 3$) in a three-dimensional input space ($n = 3$). We train the multinomial logistic regression model on this set using gradient descent.

In one of the iterations of gradient descent, the weight matrix is as follows (columns correspond to weights for the individual classes):

$$\mathbf{W} = \begin{pmatrix} 3 & 3 & 3 \\ 2 & 0 & -2 \\ 3 & -4 & 6 \\ -3 & 0 & 2 \end{pmatrix}$$

Note: The model includes a bias term, so the input vector is augmented as $x' = [1,\ x_1,\ x_2,\ x_3]^T$.

One example in the training set is $\mathbf{x} = (-4,\ -1,\ -3)$ with label $\mathbf{y} = (1, 0, 0)$. Using the augmented input vector $x' = [1,\ -4,\ -1,\ -3]^T$, what is the amount of cross-entropy loss incurred by this example in this iteration of the optimization procedure?

A. 4.02

B. 12.02

C. 6.00

D. 8.00

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Machine Learning

You have a dataset of 200 points, with 100 belonging to class -1 and 100 belonging to class $+1$. The two classes can be perfectly separated by a linear classifier, so the training error on the clean dataset is 0.

Now suppose the dataset is corrupted: 20 out of the 200 labels are flipped at random ($+1 \rightarrow -1$ and $-1 \rightarrow +1$).

The feature vectors remain unchanged; only the labels are flipped.

If we evaluate the classifier that was perfect on the clean dataset, what will its performance be on the corrupted dataset?

A. Fraction of misclassification $= \frac{1}{20}$

B. Accuracy $= 90\%$

C. Fraction of misclassification $= \frac{1}{10}$

D. Accuracy $= 80\%$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type Award: 2 Penalty: 0.67 Machine Learning

You are training a linear logistic classifier on a dataset where some of the labels have been corrupted (flipped at random).
To reduce the negative effect of corrupted labels, you decide to assign a weight $w^{(i)}$ to each data point during training.

The weighted loss function is:

$$J(\theta) = \frac{1}{n}\sum_{i=1}^{n} w^{(i)} \log\Big(1 + \exp\Big(-y^{(i)}\left(\theta^\top x^{(i)} + \theta_0\right)\Big)\Big),$$

where $w^{(i)} \geq 0$ controls the importance of each point.

Which of the following weighting schemes will work well in this setting?

A. Give equal weight $w^{(i)} = 1$ to all points.

B. For each point, look at its 5 nearest neighbors:

- If its label agrees with the majority of neighbors, set $w^{(i)} = 1$.
- If its label disagrees with the majority, set $w^{(i)} = 0$.

C. Give lower weights to points that are very far from the decision boundary (since they are already easy to classify).

D. Assign smaller weights to points that lie very close to the decision boundary, since these are more likely to be mislabeled.

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 00sec Discuss